CSL 100/ MAL 100

Tutorial 12

File handling

Q. Write a program to count the number of lines in a file data.txt.

Expected Output :

The content of the file test.txt are :

test line 1

test line 2

test line 3

test line 4

The lines in the file are : 4

```c
#include <stdio.h>

int main() {
    FILE *file = fopen("data.txt", "r");
    if (file == NULL) {
        printf("Error opening file!\n");
        return 1;
    }
    char ch;
    int lines = 0;
    while ((ch = fgetc(file)) != EOF) {
        if (ch == '\n') {
            lines++;
        }
    }
    fclose(file);
    printf("Number of lines: %d\n", lines);
    return 0;
}
```

Q. Write a program to append text "This is appended text." to an existing file data.txt.

Expected outcome
Count the number of words

Input the filename to be opened: test1.txt
The content of the file test1.txt are:
test line 1
test line 2
test line 3
test   line 4
The number of words in the file test1.txt are: 12

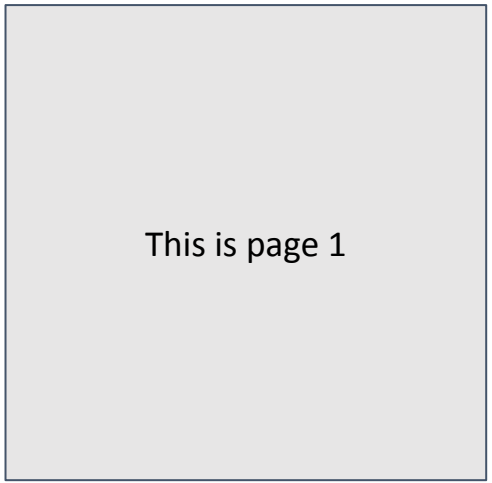
```c
#include <stdio.h>

int main() {
    FILE *file = fopen("data.txt", "a");
    if (file == NULL) {
        printf("Error opening file!\n");
        return 1;
    }
    fprintf(file, "\nThis is appended text.");
    fclose(file);
    printf("Text appended successfully.\n");
    return 0;
}
```

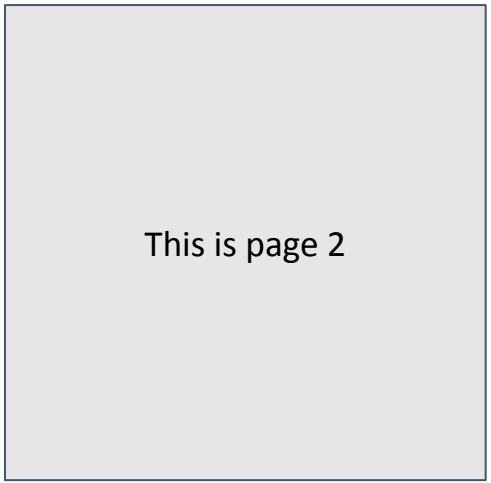
Q. Write a program to merge the contents of two files, file1.txt and file2.txt, into a new file merged.txt.

Expected output

file1.txt



file2.txt



merged.txt



```
#include <stdio.h>
void mergeFiles(FILE *file1, FILE *file2, FILE
*mergedFile) {
    char ch;
    while ((ch = fgetc(file1)) != EOF) {
        fputc(ch, mergedFile);
    }
    while ((ch = fgetc(file2)) != EOF) {
        fputc(ch, mergedFile);
    }
}

int main() {
    FILE *file1 = fopen("file1.txt", "r");
    FILE *file2 = fopen("file2.txt", "r");
    FILE *mergedFile = fopen("merged.txt", "w");
    if (file1 == NULL || file2 == NULL || mergedFile ==
NULL) {
        printf("Error opening files!\n");
        return 1;
    }
```

```
    mergeFiles(file1, file2, mergedFile);
    fclose(file1);
    fclose(file2);
    fclose(mergedFile);
    printf("Files merged successfully.\n");
    return 0;
}
```

Q. Write a program to read the last 10 characters of a file data.txt

Input the filename to be opened: test1.txt i/p-{welcome to csl100 class}
The last 10 characters of a file data.txt are l100 class

```c
#include <stdio.h>
int main() {
    FILE *file = fopen("data.txt", "r");
    if (file == NULL) {
        printf("Error opening file.\n");
        return 1;
    }
    if (fseek(file, -10, SEEK_END) != 0) {
        printf("Error seeking in file.\n");
        fclose(file);
        return 1;
    }

    char ch;
    while (fscanf(file, "%c", &ch) == 1) {
        fprintf(stdout, "%c", ch);
    }

    fclose(file);
    return 0;
}
```

1. Write a program which will search for user given word in the file and replace that word with another word given by user.

Example:
File.txt
This is csl100 / mal100 class.

Input
Enter word to search and replace:
mal100
mal400

Output:
This is csl100 / mal400 class.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
void findAndReplaceInFile()
{
   FILE *ifp, *ofp;
   char word[100], ch, read[100], replace[100];
   int word_len, i, p = 0;
   ifp = fopen("file_search_input.txt", "r");
   ofp = fopen("file_replace_output.txt", "w+");
   if (ifp == NULL || ofp == NULL) {
      printf("Can't open file.");
      exit(0);
   }
   puts("THE CONTENTS OF THE FILE:\n");
   while (1) {
      ch = fgetc(ifp);
      if (ch == EOF) {
         break;
      }
      printf("%c", ch);
   }
 puts("\n\nEnter the word to find:");
fgets(word, 100, stdin);
word[strlen(word) - 1] = word[strlen(word)];
puts("Enter the word to replace it with :");
fgets(replace, 100, stdin);
replace[strlen(replace) - 1] = replace[strlen(replace)];
 fprintf(ofp, "%s - %s\n", word, replace);
rewind(ifp);
while (!feof(ifp)) {
   fscanf(ifp, "%s", read);
   if (strcmp(read, word) == 0) {
      strcpy(read, replace);
   }
   fprintf(ofp, "%s ", read);
}
rewind(ofp);
while (1) { ch = fgetc(ofp);
   if (ch == EOF) {
      break;
   }
   printf("%c", ch);
}
  fclose(ifp);
   fclose(ofp);
}
void main()
{
findAndReplaceInFile();
}
```

QUIZ

Q. Write a C program that searches for the word help in a file. Once the word help is found, prompt the user to input another word and add that word immediately after help in the file

file.txt
This program is designed to help you understand file handling.

User Input:
Enter the file name: example.txt
Enter the word to add after 'help': students

Output
This program is designed to help students you understand file handling.